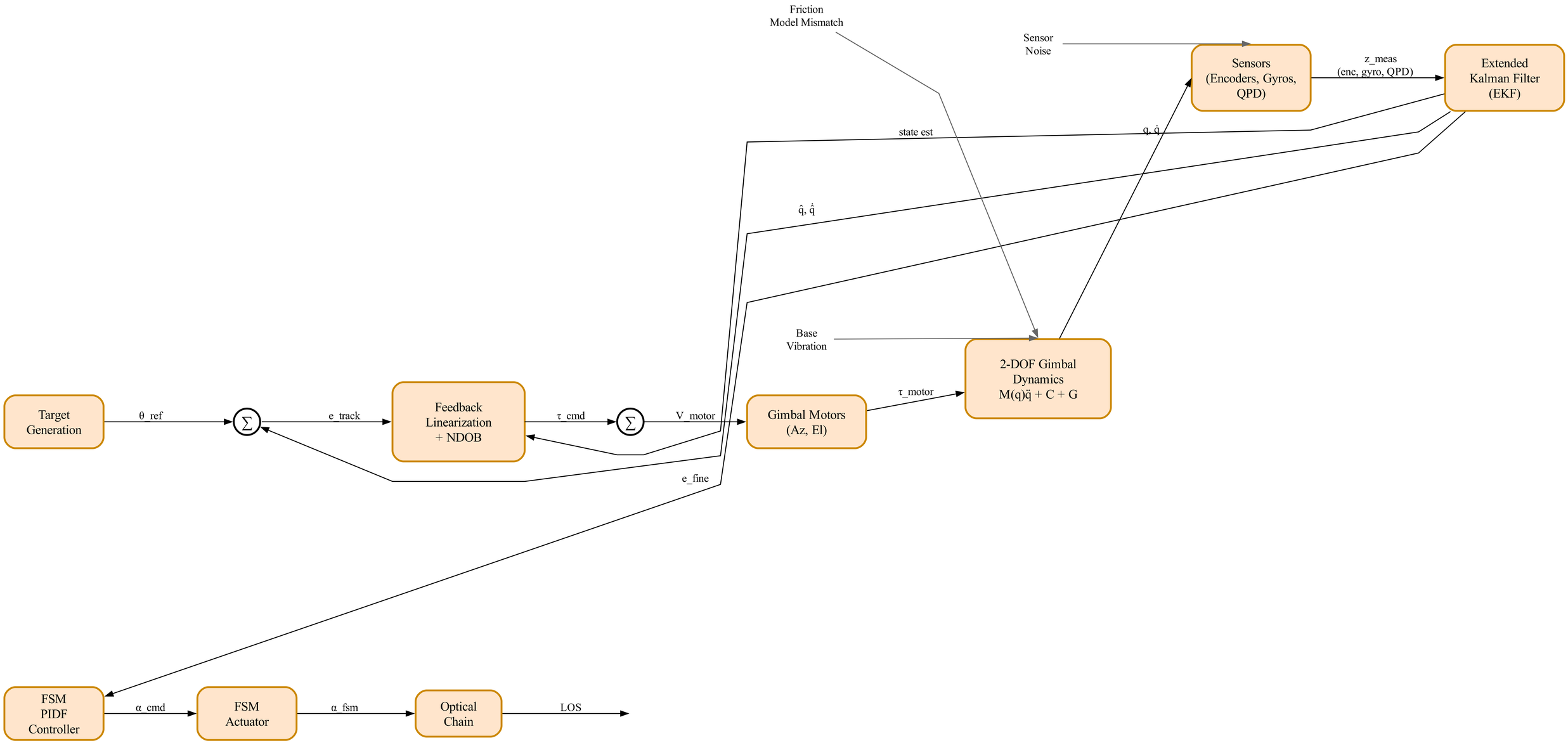
Friction
Model Mismatch
Sensor
Noise
Sensors
(Encoders, Gyros,
QPD)
z_meas
(enc, gyro, QPD)
Extended
Kalman Filter
(EKF)
state est
q, q̇
q̂, q̂̇
Base
Vibration
2-DOF Gimbal
Dynamics
M(q)q̈ + C + G
τ_motor
Target
Generation
θ_ref
Σ
e_track
Feedback
Linearization
+ NDOB
τ_cmd
Σ
V_motor
Gimbal Motors
(Az, El)
e_fine
FSM
PIDF
Controller
α_cmd
FSM
Actuator
α_fsm
Optical
Chain
LOS